Q.27 Write an algorithm for deleting element from array.

Q.28 Differentiate between primary and secondary data types.

Q.29 Differentiate between linear search and binary search.

Q.30 Write an algorithm for inserting a node in a binary search tree.

Q.31 Write a note on

a) Trees b) De-queues.

Q.32 Write a short note on structured programming.

Q.33 Explain the transversal of linear list with its algorithm.

Q.34 Explain recursion with example.

Q.35 Create an ordered binary tree for data 15, 19, 6, 13,9,17, 4, 12

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain Array. list various operations that can be performed on Array's:

Q.37 Write a note on

a) Linked list

b) Doubly linked lists

Q.38 Explain applications of stack and queue in real life.

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4th Sem / Comp
Subject:- Data Structures Using C

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is the first step in the program Development Life Cycle?

a) Coding b) Problem Definition

c) Debugging d) Documentation

Q.2 In C, a variable must be:

a) Declared before use

b) Used before declaration

c) Automatically created

d) Not declared

Q.3 A pointer in C is used to store:

a) Constant value b) Address of a variable

c) Array index d) None of the above

Q.4 Which data structure follows LIFO (Last In first Out)?

a) Queue b) Linked list

c) Array d) Stack

Q.5 Which searching algorithm is best for unsorted data?

a) Binary Search b) Hashing

c) Linear Search d) None

Q.6 Which of the following is a linear data structure?

a) Array b) Stack

c) Queue d) All of these

Q.7 A linked list is better than an array in terms of :

a) Random access

b) Dynamic memory allocation

c) Cache performance

d) Indexing

Q.8 The depth of a tree is equal to:

a) Longest path from root to leaf

b) Number of nodes

c) Number of edges

d) Height of root

Q.9 Which of the following is a non-linear data structure.

a) Trees b) Linked List

c) Queue d) All of the above

Q.10 Which of the following is a divide-and-conquer algorithm?

a) Bubble Sort b) Quick Sort

c) Selection Sort d) Insertion Sort

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define algorithm.

Q.12 What is Identifier.

Q.13 Define array.

Q.14 What is top of stack?

Q.15 ______ variable are automatically initialized to zero.

Q.16 List one type of sorting algorithm.

Q.17 What is malloc ( )?

Q.18 Define variable.

Q.19 Give an example of one-way queue.

Q.20 Write the name of two operation that can be applied on tree.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Differentiate between top-down and bottom-up approach.

Q.22 Differentiate between array and linked list.

Q.23 Explain operations on queue with example.

Q.24 Differentiate between circular and non-circular linked list.

Q.25 Explain quick sort algorithm.

Q.26 Differentiate between tree and graph.